



Solving vertex coloring problems as maximum weight stable set problems



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ARTICLE INFO

Article history:

Received 19 June 2015

Received in revised form 20 July 2016

Accepted 4 September 2016

Available online 29 September 2016

Keywords:

Vertex coloring

Stable set

Mixed integer linear programming

Computational experiments

ABSTRACT

In Vertex Coloring Problems, one is required to assign a color to each vertex of an undirected graph in such a way that adjacent vertices receive different colors, and the objective is to minimize the cost of the used colors. In this work we solve four different coloring problems formulated as Maximum Weight Stable Set Problems on an associated graph. We exploit the transformation proposed by Cornaz and Jost (2008), where given a graph G , an auxiliary graph $\hat{G}$ is constructed, such that the family of all stable sets of $\hat{G}$ is in one-to-one correspondence with the family of all feasible colorings of G . The transformation in Cornaz and Jost (2008) was originally proposed for the classical Vertex Coloring and the Max-Coloring problems; we extend it to the Equitable Coloring Problem and the Bin Packing Problem with Conflicts. We discuss the relation between the Maximum Weight Stable formulation and a polynomial-size formulation for the VCP, proposed by Campêlo et al. (2008) and called the *Representative formulation*. We report extensive computational experiments on benchmark instances of the four problems, and compare the solution method with the state-of-the-art algorithms. By exploiting the proposed method, we largely outperform the state-of-the-art algorithm for the Max-coloring Problem, and we are able to solve, for the first time to proven optimality, 14 Max-coloring and 2 Equitable Coloring instances.

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1. Introduction

In Vertex Coloring Problems, one is required to assign a color to each vertex of an undirected graph in such a way that adjacent vertices receive different colors, and the objective is to minimize the cost of the used colors. In the classical Vertex Coloring Problem (VCP), all colors have the same cost, hence, the objective is to minimize the number of used colors. The Equitable Coloring Problem is a VCP with the additional restriction that subsets of vertices receiving the same color, denoted as *color classes*, differ in cardinality of at most one unit. The Max-coloring Problem is defined as a VCP where each vertex has a positive weight, and the cost of a color is given by the maximum weight of the vertices in the corresponding color class. Finally, by considering a VCP with a positive weight associated with each vertex, and imposing that the total weight of the vertices in a color class does not exceed a given capacity, we obtain a VCP with capacity constraints. Because the problem generalizes the Bin Packing Problem as well, it is known in the literature as Bin Packing Problem with Conflicts. All mentioned problems are NP-hard.

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Coloring problems are very challenging from the computational viewpoint and have several relevant applications, including, just to mention a few, scheduling [17], timetabling [8], frequency assignment [14], register allocation [5] and communication networks [27] (see Malaguti and Toth, [20]).

Problems definition. Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. A stable set $S \subseteq V(G)$ is a subset of vertices containing no edge, a clique $K \subseteq V(G)$ is a subset of vertices inducing a complete subgraph. The stable sets of G are the cliques of the complementary graph $\bar{G}$ of G .

Given a weight $c \in \mathbb{Z}^{V(G)}$, the *Max Weight Stable Set Problem* (MWSSP) is to determine a stable set S of G maximizing $\sum_{v \in S} c(v)$. We denote by $\alpha(G, c)$ the optimum of MWSSP. The MWSSP can be naturally formulated as a (Mixed-)Integer Linear Program (MIP) since $\alpha(G, c) = \max c^\top x$ over x in $STAB(G)$, that is, the set of vectors of $\mathbb{R}^{V(G)}$ satisfying

$$\begin{cases} x_u + x_v \leq 1 & uv \in E(G) \\ x_v \in \{0, 1\} & v \in V(G). \end{cases}$$

A p -coloring of G is a partition $\mathcal{S} = S_1, \dots, S_p$ of $V(G)$ into p stable sets S_i , where each stable set represents a color. The *Vertex Coloring Problem* (VCP) is to find a p -coloring of G with a minimum number of colors p . We denote by $\chi(G)$ the optimum of VCP. The VCP is a very challenging problem from the computational viewpoint, for which state-of-the-art exact methods can fail in optimally solving instances with more than 200 vertices. The best performing exact methods for the VCP are based on the Set Covering formulation of the problem, where binary variables are associated with stable sets of G . Since stable sets of an arbitrary graph G are in exponential number with respect to the graph size, Set Covering formulations require column generation techniques and Branch-and-Price algorithms for managing the exponentially many variables. The recent algorithms by Malaguti, Monaci and Toth [19], Gualandi and Malucelli [15], and Held, Cook and Sewell [16], are all very sophisticated implementations of a Branch-and-Price algorithm, embedding specialized (meta)heuristic procedures [19], constraint programming techniques [15] and improved algorithms for the column generation subproblem [16]. For random graphs, competitive experimental results are obtained by implicit enumeration schemes based on the DSATUR algorithm by Brélaz [3], see, e.g., [26]. We also mention the Branch-and-Cut approach by Méndez-Díaz and Zabala [23], which is effective for some special classes of graphs.

Given a graph G , a coloring $\mathcal{S} = S_1, \dots, S_p$ is *equitable* if $|S_i| - |S_j| \leq 1$ for each $i, j = 1, \dots, p$. The *Equitable Coloring Problem* (ECP) is to find an equitable coloring with minimum p . We denote by $\chi_{eq}(G)$ the optimum of ECP. The most recent mathematical programming contributions to the exact solution of the ECP include the Branch-and-Cut algorithm by Bahiense et al. [1], which exploits an asymmetric formulation; and the MIP formulation by Méndez-Díaz, Nasini and Severín [22], which is strengthened by valid inequalities derived from a polyhedral study, and can be solved directly by a MIP solver. In addition, a DSATUR based Branch-and-Bound algorithm was recently proposed by Méndez-Díaz, Nasini and Severín [21]. To the best of our knowledge, no extended formulation with variables associated with stable sets of G was proposed. A possible reason is the non straightforward definition of equitable cardinality constraints for the color classes in this setting.

Given a graph G and vertex weights $c \in \mathbb{Z}^{V(G)}$, the *Max-coloring Problem* (also known as *Weighted VCP*, see [18,13]) is the problem of determining a coloring $\mathcal{S} = S_1, \dots, S_p$ of G which minimizes $\psi(\mathcal{S}) := \sum_{i=1}^p c_i$ where $c_i = \max_{v \in S_i} c(v)$. We denote by $\chi_{max}(G, c)$ the optimum of Max-col. When c is a unit vector, $\psi(\mathcal{S}) = p$ and Max-col reduces to the VCP. The best performing exact method for the Max-col is the Branch-and-Price algorithm by Furini and Malaguti [13], which can solve instances with up to 100 vertices.

Given a graph G , vertex weights $c \in \mathbb{Z}^{V(G)}$, and a nonnegative integer κ , the *Bin Packing Problem with Conflict* (BPPC) is to determine a coloring $\mathcal{S} = S_1, \dots, S_p$ of G which minimizes p so that the weight of the stable sets $c(S_i) := \sum_{v \in S_i} c(v)$ does not exceed κ . The optimum is denoted by $\chi_{bp}(G, c, \kappa)$. State-of-the-art algorithms for the BPPC, recently proposed by Fernandes-Muritiba et al. [11], Elhedhli et al. [9], Sadykov and Vanderbeck [25], exploit a Set Covering formulation and implement a Branch-and-Price framework. Apparently, the presence of capacity constraints on the cardinality of the color classes reduces the practical difficulty of solving the BPPC. The mentioned algorithms can solve to optimality instances with up to 500 vertices.

In this work we solve the VCP, ECP, Max-col and BPPC by reformulating them as MWSSPs on an associated graph. We exploit the transformation proposed by Cornaz and Jost [6], where given a graph G , an auxiliary graph $\hat{G}$ is constructed, such that the family of all stable sets of $\hat{G}$ is in one-to-one correspondence with the family of all feasible colorings of G . The transformation was originally proposed for the VCP and for the Max-col. We extend the transformation to the ECP and the BPPC; in these cases additional constraints have to be defined for the MWSSP. The advantage of this approach relies on simplicity: it allows us to solve coloring problems by solving MWSSPs, which can be tackled by problem specific algorithms, or can be formulated as MIPs of polynomial size, and solved by a general purpose MIP solver. We discuss the relation between the MWSSPs reformulation and a polynomial-size formulation for the VCP, proposed by Campêlo, Corrêa and Campos [4] and called the *Representative formulation*. To the best of our knowledge the computational performance of this latter formulation has not been investigated in the literature. Instead, computational results are presented by Bahiense et al. [1], where a Branch-and-Cut algorithm based on the Representative formulation is developed for the Equitable Coloring problem.

The transformation by Cornaz and Jost [6] was exploited by Bonomo, Giandomenico and Rossi to derive polynomial-time algorithms for special classes of graphs [2]. Recently, the same transformation was exploited by Furini, Gabrel and Ternier [12] to derive strong and fast lower bounds for the VCP, which are then used to design a DSATUR based Branch-and-Bound algorithm. The transformation was also exploited in [7] to improve the Theta-Lovász lower bound for VCP.

Given the transformation we exploit, where $\hat{G}$ has one vertex for each edge in the complement graph of G , the method is notably promising for instances where G is dense, thus producing auxiliary graphs $\hat{G}$ of manageable size.

2. “Representative” or “Sandwich graph” formulation

Let us recall the Representative formulation introduced in [4] for the VCP. Observe that any p -coloring $S_1, \dots, S_p$ of any given graph $G = (V, E)$ is also a partition into p cliques of its complementary graph $\bar{G} = (V, \bar{E})$. Assume that $\vec{G} = (V, \vec{E})$ is an orientation of $\bar{G}$ so that every clique of $\bar{G}$ is acyclic. For instance, $\vec{G}$ could be an acyclic orientation but it is not strictly necessary, a necessary and sufficient condition is that $\vec{G}$ has no 3-dicycle. (Indeed, if a complete graph has a orientation with dicycles, the minimum length of a dicycle is necessarily three.)

So, thanks to the way the orientation was chosen, in every (nonempty) clique $K \subseteq V$ of $\vec{G}$ there is a unique vertex $v \in K$ which is not the head of any arc in K . We call v the *representative* of K . Moreover, there is a unique star $\vec{S} \subseteq \vec{E}$, composed of the $|K| - 1$ arcs having tail v and whose head is a vertex in $K \setminus \{v\}$. So in $\vec{G}$, any clique K has both a unique representative v , and a unique star $\vec{S}$ (which may be empty if K is a 1-clique). A *representative set* of G , is a subset of vertices $v_1, \dots, v_p$ so that there is a clique partition $K_1, \dots, K_p$ in $\vec{G}$, where v_i is the representative of K_i .

Thus, any given p -coloring $S_1, \dots, S_p$ of G is also a clique partition in $\vec{G}$, which has a unique representative set $v_1, \dots, v_p$, and uniquely defined stars $\vec{S}_1, \dots, \vec{S}_p$ so that any vertex is either a representative or a leaf of some star. This way, finding the chromatic number of G amounts to find a representative set with minimum cardinality. This last problem can be formulated in the space $(x, y) \in \mathbb{R}^{V \times \vec{E}}$ as follows

$$\chi(G) = \min \sum_{v \in V} y_v \quad \text{s.t.} \quad (1)$$

$$y_v + \sum_{e \in \delta^-(v)} x_e = 1 \quad v \in V \quad (2)$$

$$x_{vw} \leq y_v \quad v \in V, vw \in \vec{E} \quad (3)$$

$$x_{vu} + x_{vw} \leq y_v \quad v, u, w \in V : vu, vw \in \vec{E}, uw, wu \notin \vec{E} \quad (4)$$

$$y \in \{0, 1\}^V, x \in \{0, 1\}^{\vec{E}}, \quad (5)$$

where $\delta^-(v)$ denotes the set of arcs in $\vec{E}$ with head v . In the above formulation, y is the characteristic vector of a representative set $v_1, \dots, v_p$, and x is the characteristic vector of the corresponding stars $\vec{S}_1, \dots, \vec{S}_p$ of $\vec{G}$. Constraints (2) and (3) ensure that each vertex is either a representative or the leaf of some star, and constraints (4) ensure that every star spans a stable set of G (clique of $\bar{G}$). Notice that we can project out the y variables using Fourier–Motzkin elimination, and we obtain

$$\chi(G) = |V| - \max \sum_{e \in \vec{E}} x_e \quad \text{s.t.} \quad (6)$$

$$x_{vw} + \sum_{e \in \delta^-(v)} x_e \leq 1 \quad v \in V, vw \in \vec{E} \quad (7)$$

$$x_{vu} + x_{vw} + \sum_{e \in \delta^-(v)} x_e \leq 1 \quad v, u, w \in V : vu, vw \in \vec{E}, uw, wu \notin \vec{E} \quad (8)$$

$$x \in \{0, 1\}^{\vec{E}}. \quad (9)$$

One can observe that the feasible vectors x are in actual fact the characteristic vectors of the stable sets of an auxiliary graph $\hat{G}$ of $\vec{G}$ described by Cornaz and Jost [6]. In order to build $\hat{G}$, consider the line-graph $L(\bar{G})$ of the complementary graph $\bar{G}$ of G , that is, $L(\bar{G})$ has vertex-set the edge-set of $\bar{G}$ and two vertices are linked in $L(\bar{G})$ if the two corresponding edges are adjacent in $\bar{G}$. The auxiliary graph $\hat{G}$ is obtained from $L(\bar{G})$ by removing each edge of $L(\bar{G})$ corresponding to a simplicial pair of arcs of $\vec{G}$, that is, two arcs with the same tail and with their heads linked by another arc.

In other words, the vectors x satisfying (7)–(9) are exactly the vector x satisfying (11)–(14) and the following MIP is a formulation for the VCP:

$$\max \sum_{e \in \vec{E}} x_e \quad \text{s.t.} \quad (10)$$

$$x_{vu} + x_{vw} \leq 1 \quad v, u, w \in V : vu, vw \in \vec{E}, uw, wu \notin \vec{E} \quad (11)$$

$$x_e + x_f \leq 1 \quad v \in V, e \in \delta^-(v), f \in \delta^+(v) \quad (12)$$

$$x_e + x_f \leq 1 \quad v \in V, e, f \in \delta^-(v) \quad (13)$$

$$x \in \{0, 1\}^{\vec{E}} \quad (14)$$

where we denote by $\delta^+(v)$ the set of arcs in $\vec{E}$ with tail v . Constraints (11)–(14) define $\text{STAB}(\hat{G})$, where $\hat{G}$ is the auxiliary graph defined above. For Max-col, weights $\hat{c}_e$ for arcs $e \in \vec{E}$ are introduced and the problem becomes a MWSSP.

In [6] it is also explained how to transform a weighted graph (G, c) to an auxiliary weighted graph $(\hat{G}, \hat{c})$ so that:

Theorem 1 ([6]). $\alpha(\hat{G}, \hat{c}) + \chi_{\max}(G, c) = \mathbf{1}^\top c$.

The first step is to order the vertices $v_1, \dots, v_n$ of V so that $i < j$ if and only if $c(v_i) \geq c(v_j)$. (The order is uniquely defined if and only if all the weights are different.) Now we let $\vec{G}$ be the corresponding acyclic orientation of $\bar{G}$, that is, each edge uv of $\bar{G}$ is orientated from u to v whenever $u = v_i$ and $v = v_j$ with $i < j$. (Hence, $\vec{G}$ is not necessarily uniquely defined if some vertices have the same weight.) Each edge of $L(\bar{G})$ corresponds to a pair of incident arcs of $\vec{G}$. The weight $\hat{c}_e$ of a vertex e of $\hat{G}$ (arc of $\vec{G}$) is c_v where v is the head of $e = uv$ (i.e., the vertex of minimum weight).

A *simplicial star* of $\vec{G}$ is a subset of arcs all having the same tail and whose heads are pairwise linked. For any coloring $S_1, \dots, S_p$ of G , each S_i is a clique of $\bar{G}$, and, since the orientation of $\bar{G}$ is acyclic, any of these cliques S_i has a unique representative v_i , which is the center of a simplicial star $\vec{S}_i$ of $\vec{G}$ which spans S_i . Remark that

$$|S_i| - |\vec{S}_i| = 1. \quad (15)$$

Moreover,

$$\sum_{v \in S_i} c_v - \sum_{e \in \vec{S}_i} \hat{c}_e = c_{v_i} \quad \text{and} \quad c_{v_i} = \max_{v \in S_i} c_v. \quad (16)$$

A *simplicial stellar forest* of $\vec{G}$ is a subset of arcs of $\vec{G}$ which can be partitioned into vertex-disjoint simplicial stars. Observe that given a simplicial stellar forest $\vec{F} = \vec{S}_1 \cup \dots \cup \vec{S}_p$ of $\vec{G}$, this corresponds to a stable set in $\hat{G}$. Since the converse holds as well, it follows that there is a 1-to-1 correspondence between the stable sets of $\hat{G}$ and the clique partition of $\bar{G}$, and hence with the colorings of G . In the following, we let $v_1, \dots, v_p$ be the centers of the simplicial stars of $\vec{G}$ (it is a representative set), and we will use the following

$$\text{If } v \in \{v_1, \dots, v_p\}, \text{ then } \delta^-(v) \cap \vec{F} = \emptyset, \text{ otherwise } |\delta^-(v) \cap \vec{F}| = 1 \quad (17)$$

where $\delta^-(v)$ is the subset of $\vec{E}$ the head of which is v , and an isolated vertex is the center of an empty star. Similarly, $\delta^+(v)$ is the set of all arcs of $\vec{G}$ the tail of which is v .

Theorem 1 follows from (16) by summing over all centers v_i . It allows to formulate any instance of Max-col as an instance of MWSSP.

In Figs. 1 and 2, we depict an example of the transformation. On the left of Fig. 1 we report a graph G with 9 vertices and 24 edges. For each vertex v we report the weight c_v . On the right of the figure we report the oriented complement graph $\vec{G}$. The simplicial stellar forest $\vec{F}$ is depicted by bold arrows. On the left of Fig. 2, we report the line graph $L(\bar{G})$, where each vertex corresponds to a pair of incident arcs of $\vec{G}$ (not depending on the orientation). On the right of the figure, we depict the auxiliary graph $\hat{G}$, where edges corresponding to simplicial pairs in $\vec{G}$ (i.e., $(v_9, v_1)(v_9, v_2)$ and $(v_6, v_9)(v_6, v_5)$) are removed. For each vertex u, v we report the corresponding weight $c_{u,v} = c_v$. Vertices in the stable set corresponding to $\vec{F}$ are represented by larger circles.

In the two following propositions, we give MIP formulations for ECP and BPPC. Both are obtained from the basic MWSSP formulation in $\hat{G}$ by adding constraints.

Proposition 1. $\chi_{\text{eq}}(G) = \min \mathbf{1}^\top x$ over all $x \in \text{STAB}(\hat{G})$ satisfying

$$x(\delta^+(u)) - x(\delta^+(v)) \leq 1 + x(\delta^-(v))(|\delta^+(u)| - 1) \quad \text{for all } u, v \in V \quad (18)$$

where $x(A)$ stands for $\sum_{a \in A} x_a$.

Proof of Proposition 1. If u is not a center, then $x(\delta^+(u)) = 0$ and the inequality is trivially satisfied. If v is not a center, then by (17), we have $x(\delta^-(v)) = 1$ and again the inequality is trivially satisfied. When both u and v are centers, the constraints ensure that $|\vec{S}_u| - |\vec{S}_v| \leq 1$ where $\vec{S}_u, \vec{S}_v$ are the simplicial stars of u, v . By (15), the coloring is 1-balanced. $\square$

Proposition 2. $\chi_{\text{bp}}(G, c, \kappa) = \min \mathbf{1}^\top x$ over all $x \in \text{STAB}(\hat{G})$ satisfying

$$c_v + \sum_{w: vw \in \delta^+(v)} c_w x_{vw} \leq \kappa \quad \text{for all vertex } v \in V. \quad (19)$$

Proof of Proposition 2. If v is not a center, then $c_v \leq \kappa$ otherwise the problem is unfeasible. If v is a center of a simplicial star $\vec{S}_v$, then the left-hand-side of the constraint is equal to the weight $c(S_v)$ of the corresponding stable set. $\square$

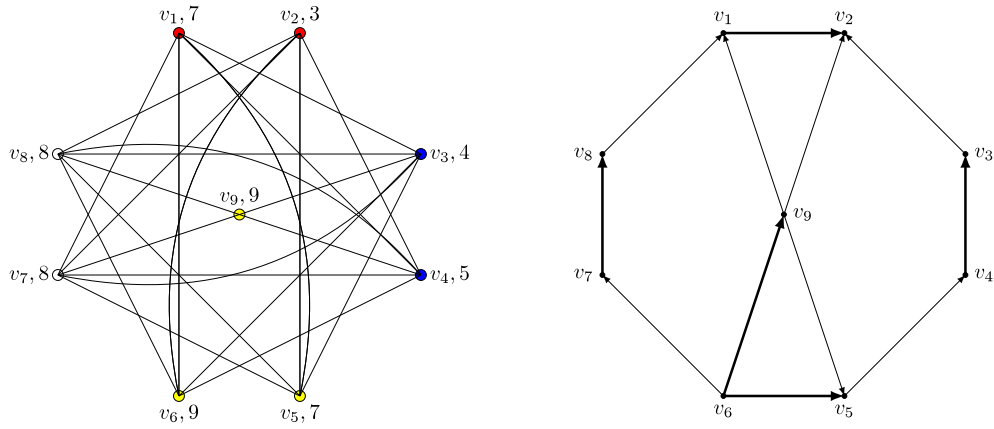


Fig. 1. Example: a graph G (left) and the orientation of its complement $\bar{G}$ (right).

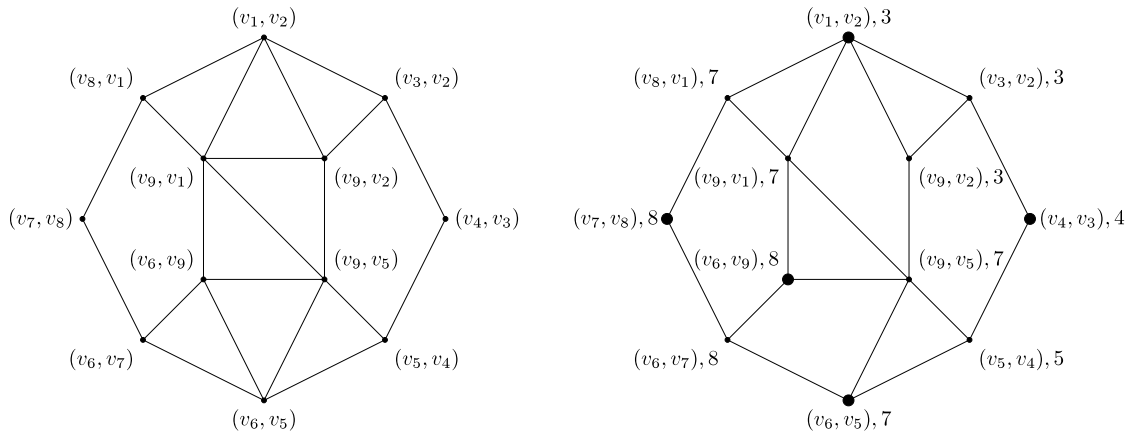


Fig. 2. Example: the line-graph $L(\bar{G})$ (left) and auxiliary graph $\hat{G}$ (right).

3. Computational experiments

We considered VCP, Max-col, ECP and BPPC instances, we derived the auxiliary graph $\hat{G} = (\hat{V}, \hat{E})$ as described in the previous section, and solved the resulting MSSP formulated as the MIP (10)–(14). For Max-col, weights $\hat{c}_e$ for arcs $e \in \vec{E}$ are considered.

Additional constraints (18) and (19) are considered in the case of the ECP and BPPC, respectively (see Section 2). In the following, we denote this solution technique by $MWSS(\hat{G})$.

In addition, we tested the Representative formulation (1)–(5), we denote this solution technique by $REPf(G)$. Both MIP formulations are solved by the general purpose commercial MIP solver CPLEX 12.6 (see also Section 3.1).

Results are compared with state-of-the-art algorithms for the corresponding problems, all based on Branch-and-Price for the VCP, Max-col and BPPC, and Branch-and-Cut algorithms and a Branch-and-Bound algorithm for the ECP. An important parameter in these experiments is the *density* of the graph, denoted by δ , which is the number of edges divided by the maximum possible number of edges (so $\delta(G) + \delta(\bar{G}) = 1$).

3.1. Algorithms for the maximum weight stable set problem

MWSSPs can be solved by means of specialized algorithms, or formulated as a MIP (see Section 1) and directly tackled by a MIP solver.

In our preliminary experiments we considered the following algorithms for solving MWSSPs: (i) the Cliquer algorithm by Ostergård, [24]; (ii) the Combinatorial Branch and Bound algorithm by Held et al. [16]; (iii) the MIP formulation of the MWSSP solved by CPLEX 12.6.

According to our experiments, the best choice is to tackle directly the MIP formulation of the MWSSP by CPLEX. This result is partially in contrast to what could be expected, since specialized algorithms are known to be faster when a MWSSP has to be solved as a column generation subproblem in a Branch-and-Price algorithm (see, e.g., [19,16,13]). Note that in the column

generation case, the MWSSP to be solved are defined on smaller-size graphs. From our computational experience, for very large graphs, like the ones obtained from the transformation, specialized algorithms tend to be slower. For example, for the 53 VCP instances considered in the next section, within a time limit of 1 h of computing time, CPLEX was able to solve to optimality the MWSSP on graph $\hat{G}$ for 38 instances, versus the 3 and 6 instances solved to optimality by Cliquer [24] and the Branch-and-Bound algorithm in [16], respectively. In the following we report results obtained by solving the MIP formulation of the MWSSP (with additional constraints (18) and (18) in the case of the ECP and BPPC, respectively) by CPLEX 12.6.

3.2. Effect of the orientation of the complement graph $\bar{G}$

The orientation $\vec{E}$ of the complement graph $\bar{G}$ has a strong effect on the computing time of the $MWSS(\hat{G})$ and $REPf(G)$ solution procedures, when the VCP, ECP and BPPC are considered. We observed that, compared with a random acyclic orientation, improved results are obtained when arcs in $\vec{E}$ are oriented in such a way that the degree (number of adjacent vertices) of the tail is not smaller than the degree of the head. Indeed, we experimentally observed that this orientation produces an auxiliary graph $\hat{G}$ containing larger cliques, and hence the $MWSS(\hat{G})$ has to solve a stable set problem which tends to be less challenging for a MIP solver. Similarly, in the $REPf(G)$, if the first k vertices of the ordering define a clique, it is straightforward to show that the linear programming relaxation of model (1)–(5) has at least value k . Ordering the vertices by non-increasing degree has the effect of producing a MIP with a stronger linear programming relaxation. In the case of the VCP, the overall computing time on the common subset of solved instances is largely reduced (approximately 80%) for both the $MWSS(\hat{G})$ and the $REPf(G)$; the $MWSS(\hat{G})$ can solve 3 and $REPf(G)$ can solve 4 more instances to optimality within time limit, respectively. In the next sections we report results obtained according to this ordering. In the case of the Max-col, the orientation is defined by the relative weight of graph G vertices.

3.3. Computational results for the VCP

None of the Branch-and-Price algorithms [19,15] and [16] clearly dominates the others, hence, we compare our results with [19], which reports an extensive set of computational results for benchmark instances. Among the 115 DIMACS benchmark instances¹ considered in [19], we restricted the set to 58 instances for which the auxiliary graph $\hat{G}$ has at most 30,000 vertices. In Table 1 we report the cardinality of the vertex sets $|V(G)|$ and $|V(\hat{G})|$, and of the edge sets $|E(G)|$ and $|E(\hat{G})|$, respectively, for the original and transformed graphs. Clearly, the size of $V(\hat{G})$ remains limited for dense graphs G , which are the instances where the method we study is more promising. To allow a fair comparison on results obtained with different machines, a benchmark program (dfmax) together with a benchmark instance (r500.5) is available. Computing times obtained on different machines can be scaled with respect to the performance obtained on this program. In our experiments, we scale time limits according to the relative speed of the computer we use, but we report original computing times from the respective papers.

In the table we then report the results obtained by the $MWSS(\hat{G})$ and $REPf(G)$ with a limit of 6.5 h of computing time, on a single node of a cluster containing machines equipped with an Intel Xeon E3-1220 processor at 3.10 GHz and 8 GB RAM. This machine needs 4.5 s to run the benchmark program on the instance. We report, for each considered instance and each method, a lower bound LB and an upper bound UB on the value $\chi(G)$, and the corresponding computing time (tl when time limit is reached), which, for $MWSS(\hat{G})$, includes the (very small) time for applying the transformation of G to $\hat{G}$. In the next columns, we report the results of the Branch-and-Price algorithm in [19] (columns BP [19]), obtained within 10 h of computing time on a Pentium 4 PC with 2 GB RAM, which needs 7 s to solve the benchmark program on the instance.

On the considered 58 instances, the $REPf(G)$ method performs very well solving to optimality 41 instances. The $MWSS(\hat{G})$ method can solve 39 instances, with comparable computing times with some notable exception, for instance DSJC125.1 and instances of the Insertion and FullIns sets, where the computing times of the $REPf(G)$ are shorter for solved problems. Despite one would expect CPLEX MIP solver to be more effective in solving models formulated as (unweighted) stable set problems, our computational tests showed that instead this is not the case. In particular, we observed that the LP solver of CPLEX tends to spend more time when solving the linear programming relaxation of the $MWSS(\hat{G})$ than the $REPf(G)$. Both methods perform very well when compared with the 39 instances optimally solved by [19]. In particular $MWSS(\hat{G})$ can solve in less than one hour instance DSJC250.9, which was only recently solved to optimality by [16] with almost 3 h of computing time.

3.4. Computational results for the Max-col

The best performing algorithm for Max-col, i.e., the Branch-and-Price algorithm [13], can solve instances with up to 100 vertices. This makes the $MWSS(\hat{G})$ method an excellent candidate alternative for solving the Max-col, since we expect the weighted auxiliary graph $\hat{G}$ being not too large, and we do not have to spoil the MWSSP formulation with additional constraints. We considered all the instances in [13]. This includes 46 instances from the COLOR02/03/04 benchmark² and two

¹ <ftp://dimacs.rutgers.edu/pub/challenge/graph/>.

² <http://mat.gsia.cmu.edu/COLOR02>.

Table 1
Results for the VCP, DIMACS benchmark instances.

Instance	G		$\hat{G}$		$MWSS(\hat{G})$			$REPf(G)$			BP [19]		
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	LB	UB	Time	LB	UB	Time	LB	UB	Time
DSJC125.1	125	736	7,014	545,250	5	5	17 314.7	5	5	65.5	5	5	142.0
DSJC125.5	125	3,891	3,859	197,166	14	20	tl	14	20	tl	17	17	18,290.0
DSJC125.9	125	6,961	789	9,503	44	44	0.5	44	44	0.5	44	44	3916.0
DSJC250.1	250	3,218	27,907	4,350,878	5	17	tl	5	10	tl	6	8	tl
DSJC250.5	250	15,668	15,457	1,588,249	17	48	tl	18	40	tl	20	28	tl
DSJC250.9	250	27,897	3,228	79,902	72	72	1935.2	72	72	2401.5	71	72	tl
DSJC500.9	500	112,437	12,313	584,336	123	132	tl	123	133	tl	123	127	tl
DSJR500.1c	500	121,275	3,475	39,563	85	85	1.3	85	85	1.3	85	85	288.5
queen5_5	25	160	140	1,240	5	5	0	5	5	0	5	5	0.2
queen6_6	36	290	340	5,092	7	7	0.5	7	7	0.5	7	7	6.0
queen7_7	49	476	700	15,736	7	7	0.3	7	7	0.2	7	7	0.2
queen8_8	64	728	1,288	40,344	9	9	150.6	9	9	34.9	9	9	3.6
queen9_9	81	1,056	2,184	90,672	9	10	tl	9	10	tl	10	10	36.6
queen10_10	100	1,470	3,480	184,600	11	12	tl	10	12	tl	11	11	687.0
queen11_11	121	1,980	5,280	348,040	11	13	tl	11	13	tl	12	12	1875.7
queen12_12	144	2,596	7,700	616,940	12	14	tl	12	15	tl	12	13	tl
queen13_13	169	3,328	10,868	1,039,688	13	16	tl	13	16	tl	13	14	tl
queen14_14	196	4,186	14,924	1,679,580	14	17	tl	14	18	tl	14	15	tl
queen15_15	225	5,180	20,020	2,617,720	16	27	tl	15	19	tl	15	15	tl
queen16_16	256	6,320	26,320	3,955,952		256	tl	16	24	tl	16	17	tl
myciel3	11	20	35	150	4	4	0	4	4	0	4	4	4.7
myciel4	23	71	182	2,051	5	5	0.2	5	5	0.2	5	5	123.0
myciel5	47	236	845	21,810	6	6	150.6	6	6	61.2	4	6	tl
myciel6	95	755	3,710	206,615	6	7	tl	6	7	tl	4	7	tl
myciel7	191	2,360	15,785	1,840,230	4	8	tl	4	8	tl	5	8	tl
multsol.i.1	197	3,925	15,381	1,805,912	49	49	6.8	49	49	2	49	49	0.2
multsol.i.2	188	3,885	13,693	1,498,197	31	31	6.2	31	31	12.3	31	31	4.7
multsol.i.3	184	3,916	12,920	1,371,624	31	31	4.9	31	31	9.8	31	31	0.2
multsol.i.4	185	3,946	13,074	1,396,293	31	31	5.8	31	31	10.4	31	31	0.2
multsol.i.5	186	3,973	13,232	1,421,732	31	31	5.4	31	31	11.1	31	31	6.0
zeroin.i.1	211	4,100	18,055	2,303,047	49	49	9.1	49	49	10	49	49	3.8
zeroin.i.2	211	3,541	18,614	2,383,090	30	30	11	30	30	0.5	30	30	4.4
zeroin.i.3	206	3,540	17,575	2,185,681	30	30	11	30	30	9.4	30	30	4.5
anna	138	493	8,960	790,966	11	11	2.8	11	11	0.2	11	11	3.6
david	87	406	3,335	178,437	11	11	0.4	11	11	0.1	11	11	0.2
huck	74	301	2,400	108,649	11	11	0.2	11	11	0.1	11	11	0.2
jean	80	254	2,906	144,975	10	10	0.4	10	10	0.1	10	10	0.2
games120	120	638	6,502	487,295	9	9	4.9	9	9	0.4	9	9	0.2
miles250	128	387	7,741	634,554	8	8	1.8	8	8	0.2	8	8	5.0
miles500	128	1,170	6,958	541,403	20	20	1.3	20	20	0.2	19	20	3.7
miles750	128	2,113	6,015	436,137	31	31	0.9	31	31	2	31	31	0.2
miles1000	128	3,216	4,912	323,889	42	42	0.7	42	42	3.3	42	42	0.2
miles1500	128	5,198	2,930	158,110	73	73	0.2	73	73	1	73	73	0.1
1-Insertions_4	67	232	1,979	80,730	4	5	tl	5	5	15 050.7	3	5	tl
1-Insertions_5	202	1,227	19,074	2,461,400	3	6	tl	4	6	tl	3	6	tl
2-Insertions_4	149	541	10,485	1,001,021	4	5	tl	4	5	tl	3	5	tl
4-Insertions_3	79	156	2,925	146,146	3	4	tl	4	4	13 564.5	3	4	tl
1-FullIns_4	93	593	3,685	205,688	5	5	17.3	5	5	4.8	4	5	tl
1-FullIns_5	282	3,247	36,374	6,487,330	5	6	tl	5	6	tl	4	6	tl
2-FullIns_3	52	201	1,125	34,190	5	5	0.3	5	5	0	5	5	2.9
2-FullIns_4	212	1,621	20,745	2,791,046	6	6	16 021.6	6	6	115.6	5	6	tl
3-FullIns_3	80	346	2,814	137,405	6	6	1.4	6	6	0.2	6	6	2.9
4-FullIns_3	114	541	5,900	420,462	7	7	4.3	7	7	0.4	7	7	3.4
5-FullIns_3	154	792	10,989	1,073,628	8	8	14.6	8	8	0.7	8	8	4.6
mug88_1	88	146	3,682	206,951	4	4	6.5	4	4	0.7	4	4	9.6
mug88_25	88	146	3,682	206,953	4	4	6.9	4	4	5.3	4	4	10.6
mug100_1	100	166	4,784	307,175	4	4	31	4	4	5.8	4	4	14.4
mug100_25	100	166	4,784	307,175	4	4	119.6	4	4	28.6	4	4	12.0

sets of instances from matrix-decomposition problems. In Tables 2, 4 and 5 we report the size of the original and transformed graphs, and the results obtained by the presented method (columns $MWSS(\hat{G})$), followed by the results from [13], obtained on the same computer (an i5 Linux PC with 8 GB RAM). Time limits are set to 1 h of computing time for the first to two sets (DIMACS and p) and to 4 h for the last set (R), as in [13]. We also tested the $REPf(G)$ but, for the Max-col, the $MWSS(\hat{G})$ has a slightly better performance, being able to solve one more instance to proven optimality, with similar computing times on the other solved instances. In contrast with what we have experienced for the VCP, CPLEX better solves the weighted stable set formulation ($MWSS(\hat{G})$) than the weighted extension of the $REPf(G)$ to the Max-col.

Table 2

Results for the Max-col, COLOR02/03/04 instances, time limit of 1 h.

Instance	G		$\hat{G}$		MWSS($\hat{G}$)			BP [19]		
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	LB	UB	Time	LB	UB	Time
DSJC125_1g	125	736	7014	545,250	20	30	tl	21	27	tl
DSJC125_1gb	125	736	7014	545,250	76	104	tl	82	96	tl
DSJC125_5g	125	3891	3859	197,166	62	82	tl	68	82	tl
DSJC125_5gb	125	3891	3859	197,166	209	284	tl	230	273	tl
DSJC125_9g	125	6961	789	9,503	169	169	0.3	169	169	2
DSJC125_9gb	125	6961	789	9,503	604	604	0.4	604	604	5
GEOM100	100	547	4403	271,017	65	65	3.8	65	65	12
GEOM100a	100	992	3958	230,557	89	89	2.4	89	89	2
GEOM100b	100	1050	3900	225,378	32	32	3.8	32	32	1
GEOM110	110	638	5357	364,224	68	68	59.5	68	69	tl
GEOM110a	110	1207	4788	307,120	97	97	12.9	97	97	7
GEOM110b	110	1256	4739	302,283	37	37	5.0	37	37	2
GEOM120	120	773	6367	472,507	72	72	12.4	72	72	12
GEOM120a	120	1434	5706	400,114	105	105	7.0	105	105	11
GEOM120b	120	1491	5649	393,915	35	35	16.5	35	35	1
GEOM30b	30	81	354	5,923	12	12	0.0	12	12	0
GEOM40b	40	157	623	14,023	16	16	0.1	16	16	0
GEOM50b	50	249	976	27,722	18	18	0.1	18	18	0
GEOM60b	60	366	1404	48,110	23	23	0.5	23	23	0
GEOM70	70	267	2148	91,726	47	47	0.3	47	47	1
GEOM70a	70	459	1956	79,625	73	73	0.4	73	73	1
GEOM70b	70	488	1927	77,712	24	24	0.9	24	24	0
GEOM80	80	349	2811	137,702	66	66	1.1	66	66	2
GEOM80a	80	612	2548	118,686	76	76	1.1	76	76	1
GEOM80b	80	663	2497	114,959	27	27	2.5	27	27	1
GEOM90	90	441	3564	197,010	61	61	2.0	61	61	4
GEOM90a	90	789	3216	168,570	73	73	4.8	73	73	1
GEOM90b	90	860	3145	162,945	30	30	1.7	30	30	1
R100_1g	100	509	4441	273,695	19	23	tl	20	25	tl
R100_1gb	100	509	4441	273,695	76	83	tl	76	90	tl
R100_5g	100	2456	2494	102,327	53	62	tl	56	65	tl
R100_5gb	100	2456	2494	102,327	203	225	tl	212	233	tl
R100_9g	100	4438	512	5,033	141	141	0.1	141	141	0
R100_9gb	100	4438	512	5,033	518	518	0.3	517	518	5
R50_1g	50	108	1117	34,029	14	14	0.8	14	14	1
R50_1gb	50	108	1117	34,029	53	53	1.6	53	53	0
R50_5g	50	612	613	12,251	37	37	1.4	36	37	2
R50_5gb	50	612	613	12,251	135	135	1.5	131	135	22
R50_9g	50	1092	133	654	74	74	0.0	74	74	0
R50_9gb	50	1092	133	654	262	262	0.0	262	262	0
R75_1g	70	251	2164	92,467	18	18	132.8	18	18	12
R75_1gb	70	251	2164	92,467	70	70	192.2	67	70	447
R75_5g	75	1407	1368	41,115	51	51	1056.0	49	51	2136
R75_5gb	75	1407	1368	41,115	186	186	989.3	179	187	tl
R75_9g	75	2513	262	1,724	110	110	0.0	109	110	0
R75_9gb	75	2513	262	1,724	396	396	0.0	396	396	1

Table 3

Results for the Max-col, COLOR02/03/04 instances, time limit of 10 h.

Instance	G		$\hat{G}$		MWSS($\hat{G}$)		
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	LB	UB	Time
DSJC125_1g	125	736	7014	545,250	20	25	tl
DSJC125_1gb	125	736	7014	545,250	78	95	tl
DSJC125_5g	125	3891	3859	197,166	63	78	tl
DSJC125_5gb	125	3891	3859	197,166	212	263	tl
R100_1g	100	509	4441	273,695	21	21	28,788.5
R100_1gb	100	509	4441	273,695	81	81	9362.2
R100_5g	100	2456	2494	102,327	55	59	tl
R100_5gb	100	2456	2494	102,327	210	225	tl

Clearly the results obtained by the MWSS($\hat{G}$) method largely outperform those in [13] (obtained on the same computer): concerning the 46 COLOR02/03/04 benchmark, 38 instances are solved to optimality, i.e., 2 more than [13]. By allowing up to 10 h of computing time (see Table 3), the MWSS($\hat{G}$) method can additionally solve to optimality instances R100_1g and R100_1gb, and for unsolved instances, the optimality gaps are further reduced and always smaller than those obtained

Table 4
Results for the Max-col, p instances.

Instance	G		$\hat{G}$		$MWSS(\hat{G})$			BP [19]		
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	LB	UB	Time	LB	UB	Time
P06	16	38	82	608	565	565	0.0	565	565	0
P07	24	92	184	2,112	3771	3771	0.0	3771	3771	0
p08	24	92	184	2,112	4049	4049	0.0	4049	4049	0
p09	25	100	200	2,400	3388	3388	0.0	3388	3388	0
p10	16	32	88	688	3983	3983	0.0	3983	3983	0
p11	18	48	105	904	3380	3380	0.0	3380	3380	0
p12	26	90	235	3,118	657	657	0.0	657	657	0
p13	34	160	401	7,048	3220	3220	0.0	3220	3220	1
p14	31	110	355	5,899	3157	3157	0.0	3157	3157	0
p15	34	136	425	7,760	341	341	0.0	341	341	0
p16	34	134	427	7,816	2343	2343	0.0	2343	2343	0
p17	37	161	505	10,090	3281	3281	0.0	3281	3281	0
p18	35	143	452	8,523	3228	3228	0.0	3228	3228	0
p19	36	156	474	9,196	3710	3710	0.0	3710	3710	0
p20	37	142	524	10,715	1830	1830	0.0	1830	1830	1
p21	38	155	548	11,471	3660	3660	0.0	3660	3660	0
p22	38	154	549	11,498	1912	1912	0.0	1912	1912	0
p23	44	204	742	18,179	3770	3770	0.1	3770	3770	0
p24	34	104	457	8,723	661	661	0.0	661	661	0
p25	36	120	510	10,308	504	504	0.0	504	504	0
p26	37	131	535	11,086	520	520	0.0	520	520	0
p27	44	174	772	19,356	216	216	0.1	216	216	2
p28	44	164	782	19,766	1729	1729	0.1	1729	1729	0
p29	53	254	1124	34,271	3470	3470	0.1	3470	3470	0
p30	60	317	1453	50,540	4891	4891	0.2	4891	4891	1
p31	47	179	902	24,580	620	620	0.1	620	620	0
p32	51	221	1054	31,117	2480	2480	0.1	2480	2480	0
p33	56	258	1282	41,840	3018	3018	0.3	3018	3018	1
p34	74	421	2280	100,077	1980	1980	0.6	1980	1980	2
p35	86	566	3089	158,255	2140	2140	0.6	2140	2140	2
p36	101	798	4252	256,244	7210	7210	1.4	7210	7210	4
p38	87	537	3204	167,359	2130	2130	1.2	2130	2130	10
p40	86	497	3158	163,820	4984	4984	1.0	4984	4984	13
p41	116	900	5770	406,504	2688	2688	3.2	2688	2688	31
p42	138	1186	8267	698,948	2466	2466	3.2	2466	2466	171

by [13] (while the Branch-and-Price algorithm in [13] does not benefit from additional computing time). Concerning the p instances, which are all solved to optimality by both methods, the average computing time is reduced of one order of magnitude, from 6.83 to 0.35 s. Finally, for the R instances, [13] reports only 19 optimal solutions, while all the 30 instances are optimally solved by the $MWSS(\hat{G})$ method.

3.5. Computational results for the ECP

For solving the ECP, we consider the $MWSSP$ on graph $\hat{G}$ and the additional balance constraints (18). Constraints (18) are $O(n^2)$ in total, since one constraint is defined for each pair of vertices $u, v \in V$, but the constraint is active only if u, v are the centers of simplicial stars associated with two distinct colors (see Proposition 1). Hence, there is a large number of constraints which are rarely activated. In this case, better results are obtained by defining (18) as *lazy constraints* in the CPLEX MIP solver, i.e., constraints that are not added to the formulation, but checked only when an incumbent (integer) solution is found (and eventually added to the formulation when violated).

The three recent papers [22,21,1] report computational experiments on randomly generated graphs and the DIMACS benchmark instances. We consider the same random graphs used for testing the Branch-and-Bound algorithm in [21], which also summarizes results obtained by solving the compact formulation from [22], having from 60 to 120 vertices and densities $\delta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, as reported in Table 6. Each row of the table reports average results over 30 instances for the size of the original and transformed graphs. The table also reports the number of instances solved to optimality (opt) within 2 h of computing time by the approach $MWSS(\hat{G})$, and the average computing time (for solved instances only). In the table we also report the number of optimally solved instances and the average computing time for [22,21], obtained with a PC equipped with an Intel i5 CPU at 2.67 GHz. We estimate a time ratio of 14/9 between this PC and our cluster computer, and accordingly, we scaled our time limit to 4600 s. Reported computing times from other papers are not scaled.

The approach we propose can solve all instances with 60 vertices, while 17 instances with 70 vertices and $\delta = 0.3$ and 23 instances with 70 vertices and $\delta = 0.5$ are solved. When 80 or more vertices are considered, instances with $\delta = 0.1, 0.7, 0.9$ remain tractable, while instances with $\delta = 0.3, 0.5$ confirm to be difficult for the approach. Considering the same instances, complete results for the formulation in [22] are available up to 70 vertices (300 instances in total). The formulation in [22]

Table 5Results for the Max-col, R instances.

Instance	G		$\hat{G}$		$MWSS(\hat{G})$			BP [19]		
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	LB	UB	Time	LB	UB	Time
r01	144	1280	9,016	796,427	6724	6724	17.6	6724	6724	171
r02	142	1246	8,765	763,268	6771	6771	11.3	6771	6771	200
r03	139	1188	8,403	716,330	6473	6473	10.4	6473	6473	250
r04	151	1406	9,919	919,467	6342	6342	11.4	6342	6342	565
r05	142	1266	8,745	760,628	6408	6408	12.3	6408	6408	458
r06	148	1381	9,497	861,204	7550	7550	10.3	7550	7550	136
r07	141	1253	8,617	743,949	6889	6889	13.7	6889	6889	210
r08	138	1191	8,262	698,277	6057	6057	4.1	6057	6057	289
r09	129	1027	7,229	571,102	6358	6358	9.5	6358	6358	241
r10	150	1409	9,766	898,183	6508	6508	13.1	6508	6508	227
r11	208	2247	19,281	2,501,166	7654	7654	57.2	7654	7654	2973
r12	199	2055	17,646	2,189,139	7690	7690	53.7	7690	7690	3161
r13	217	2449	20,987	2,841,217	7500	7500	105.2	7500	–	tl
r14	214	2387	20,404	2,723,361	8254	8254	65.3	8254	8254	1420
r15	198	2055	17,448	2,152,308	8021	8021	25.1	8021	8021	1157
r16	188	1861	15,717	1,839,320	7755	7755	26.7	7755	7755	712
r17	213	2392	20,186	2,679,713	7979	7979	79.3	7979	7979	2257
r18	200	2079	17,821	2,221,870	7232	7232	58.2	7232	7232	2502
r19	185	1803	15,217	1,751,992	6826	6826	32.7	6826	6826	1349
r20	217	2447	20,989	2,841,629	8023	8023	104.1	8023	8023	993
r21	281	3554	35,786	6,339,798	9284	9284	390.0	–	–	tl
r22	285	3684	36,786	6,607,860	8887	8887	302.6	–	–	tl
r23	288	3732	37,596	6,827,707	9136	9136	375.3	–	–	tl
r24	269	3284	32,762	5,552,048	8464	8464	201.7	8464	–	tl
r25	266	3177	32,068	5,376,273	8426	8426	225.6	–	–	tl
r26	284	3629	36,557	6,546,162	8819	8819	439.6	–	–	tl
r27	259	3019	30,392	4,959,581	7975	7975	248.1	7975	–	tl
r28	288	3765	37,563	6,818,692	9407	9407	222.7	9407	–	tl
r29	281	3553	35,787	6,340,075	8693	8693	388.0	8693	–	tl
r30	301	4122	41,028	7,785,349	9816	9816	346.9	9816	–	tl

Table 6

Results for the ECP, random instances from [21].

δ	G		$\hat{G}$		$MWSS(\hat{G})$		MIP [22]		BB [21]	
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	opt	Time	opt	Time	opt	Time
10	60	179.1	1590.9	58,084.5	30	1.8	30	0.0	30	0.0
30	60	534.8	1235.2	38,362.7	30	106.7	30	506.0	30	0.0
50	60	883.2	886.8	21,465.4	30	140.5	19	1825.0	30	0.6
70	60	1234.6	535.4	8442.3	30	8.6	27	942.0	30	1.6
90	60	1589.3	180.7	1044.4	30	0.0	30	1.0	30	0.0
10	70	242.6	2172.4	93,040.1	30	5.8	30	0.0	30	0.0
30	70	719.1	1695.9	62,018.6	17	3135.1	15	4005.0	30	0.1
50	70	1205.8	1209.2	34,317.8	23	1705.8	0		30	6.7
70	70	1688.7	726.3	13,353.5	30	13.9	6	236.0	30	21.0
90	70	2177.5	237.5	1544.3	30	0.0	30	258.0	30	0.1
10	80	312.7	2847.3	140,007.9	30	63.7	30	1.3	30	0.0
30	80	950.5	2209.5	92,399.3	0				30	22.7
50	80	1573.0	1587.0	51,780.9	2	4482.9			28	324.0
70	80	2217.2	942.8	19,745.3	30	200.3			21	1817.0
90	80	2843.8	316.2	2385.2	30	0.1	27	659.0	30	6.6
10	90	402.1	3602.9	199,697.2	30	346.4			30	0.0
30	90	1200.1	2804.9	132,520.9	0				30	52.0
50	90	2005.4	1999.6	73,237.9	0					
70	90	2799.7	1205.3	28,736.1	30	691.3				
90	90	3598.9	406.1	3502.0	30	0.1			30	161.0
10	100	504.7	4445.3	274,114.9	30	491.6	30	15.5	30	0.0
10	120	715.8	6424.2	477,499.4	2	4489.1	15	1673.0	30	2.6

can be solved to optimality for 217 instances, while the approach we propose can solve 280 of the same instances. The Branch-and-Bound algorithm [21] performs very well on these random problems, indeed, it can solve to optimality all the 300 instances with up to 70 vertices, and 141 instances with 80 vertices, while our approach can solve 92 instances with 80 vertices. Our approach seems to have a better performance on dense graphs: it is from one to four orders of magnitude faster for instances with $\delta = 0.9$ and 70 or more vertices (both approaches solve all instances in these classes), and can solve all instances with 80 vertices and $\delta = 0.7$, while the algorithm in [21] can solve only 21 instances in this class.

Table 7
Results for the ECP, DIMACS instances.

Instance	G		$\hat{G}$		$MWSS(\hat{G})$			MIP [22]	BC [1]	BB [21]
	$ V(G) $	$ E(G) $	$ V(\hat{G}) $	$ E(\hat{G}) $	LB	UB	Time	Time	Time	Time
miles750	128	2,113	6,015	436,137	31	31	1.84	0.0	171.0	tl*
miles1000	128	3,216	4,912	323,889	42	42	1.07	0.0	267.0	tl*
miles1500	128	5,198	2,930	158,110	73	73	0.37	0.0	13.0	0.0
zeroin.i.1	211	4,100	18,055	2,303,047	49	49	23.52	0.0	50.0	0.0
zeroin.i.2	211	3,541	18,614	2,383,090	30	211	tl	2.0	510.0	tl
zeroin.i.3	206	3,540	17,575	2,185,681	30	36	tl	5.0	491.0	tl
queen6_6	36	290	340	5,092	7	7	0.82	1.0	1.0	0.0
queen7_7	49	476	700	15,736	7	7	0.32	0.0	0.0	0.0
queen8_8	64	728	1,288	40,344	9	9	3128.91	654.0	441.0	6.0
queen8_12	96	1,368	3,192	162,036	12	12	46.62	5.0		3079.0
queen9_9	81	1,056	2,184	90,672	9	11	tl	tl		475.0
queen10_10	100	1,470	3,480	184,600	10	12	tl	tl		tl
david	87	406	3,335	178,437	16	30	tl	0.0	13.0	0.0
1-FullIns_3	30	100	335	5,342	4	4	0.08	0.0	2.0	0.0
2-FullIns_3	52	201	1,125	34,190	5	5	0.48	0.0	25.0	1.0
3-FullIns_3	80	346	2,814	137,405	6	6	1.55	0.0	85.0	tl
4-FullIns_3	114	541	5,900	420,462	7	7	5.36	0.0	72.0	tl
5-FullIns_3	154	792	10,989	1,073,628	8	8	22.48	0.0	268.0	tl
1-FullIns_4	93	593	3,685	205,688	5	5	307.27	28.0		1404.0
mug88_1	88	146	3,682	206,951	4	4	12.80	1.0		109.0
mug88_25	88	146	3,682	206,953	4	4	30.61	0.0		56.0
mug100_1	100	166	4,784	307,175	4	4	125.01	1.0		4425.0
mug100_25	100	166	4,784	307,175	4	4	797.67	1.0		4978.0
multsol.i.1	197	3,925	15,381	1,805,912	49	50	tl	1.0		0.0
multsol.i.2	188	3,885	13,693	1,498,197	31	188	tl	tl		tl
1-Insertions_4	67	232	1,979	80,730	4	5	tl	tl		1055.0
2-Insertions_3	37	72	594	13,020	4	4	3.80	0.0		0.0
3-Insertions_3	56	110	1,430	49,500	4	4	425.56	8.0		1.0
4-Insertions_3	79	156	2,925	146,146	3	4	tl	836.0		1615.0
DSJC125.1	125	736	7,014	545,250	5	6	tl	214.0		0.0
DSJC125.5	125	3,891	3,859	197,166	14	22	tl	tl		tl
DSJC125.9	125	6,961	789	9,503	44	44	0.42	tl		tl
DSJC250.1	250	3,218	27,907	4,350,878		250	tl	tl		tl
DSJC250.5	250	15,668	15,457	1,588,249		250	tl	tl		tl
DSJC250.9	250	27,897	3,228	79,902	72	72	3437.65	tl		tl
myciel4	23	71	182	2,051	5	5	0.29	0.0	5.0	0.0
myciel5	47	236	845	21,810	6	6	171.40	149.0		0.0
myciel6	95	755	3,710	206,615	5	7	tl	tl		tl
flat300_20_0	300	21,375	23,475	3,021,152		300	tl	tl		tl
kneser7_2	21	105	105	770	6	6	0.25	0.0	6.0	0.0
kneser7_3	35	70	525	10,780	3	3	0.28	0.0	2.0	0.0
kneser9_4	126	315	7,560	611,940	3	3	150.08	0.0	809.0	0.0

Concerning the Branch-and-Cut algorithm in [1], it considers 5 instances per class, with up to 70 vertices. It can solve 2 instances with 60 vertices and $\delta = 0.3$, and 3 instances with 60 vertices and $\delta = 0.5$, while no instance with 70 vertices and $\delta = 0.3$ or $\delta = 0.5$ is solved within a time limit of 2 h on a 1.8 GHz PC with 1 GB RAM. This allows us to conclude that the method we propose has a better performance than [1] on randomly generated instances.

We also experimented our method on 54 DIMACS instances and 4 *kneser* graphs considered in [22], and restrict the set to 42 instances for which the auxiliary graph $\hat{G}$ has at most 30,000 vertices. A subset of these instances was considered in [1] as well. Table 7 reports the size of the original and transformed graphs, and the results obtained by the presented method (columns $MWSS(\hat{G})$), followed by the computing times from [22,1,21], from the respective papers. The approach we propose performs satisfactorily on these instances, and solves to optimality 27 out of 42 instances within 4600 s of computing time. It is worth mentioning that we are able to solve to optimality, for the first time, instances DSJC125.9 instance DSJC250.9. The Branch-and-Bound algorithm [22] is the best performing approach for this set of instances, and can solve 31 instances to optimality within 2 h of computing time on the slower computer mentioned above, hence, with comparable effort. The Branch-and-Bound algorithm [21] can solve 26 instances (the miles750 and miles1000 instances, where time limit is marked *tl**, can be solved by choosing a different strategy in the algorithm). Finally, the algorithm in [1] solves all the 19 considered instances (the other instances are not considered).

3.6. Computational results for the BPPC

State-of-the-art algorithms for the BPPC, proposed by Fernandes-Muritiba et al. [11], Elhedhli et al. [9] and Sadykov and Vanderbeck [25] are all based on the Branch-and-Price framework. All these algorithms are tested on BPPC instances with

up to 500 items (vertices), obtained by adding random incompatibility graphs with densities in the range 0.1–0.9 to 80 BPP instances from Falkenauer [10], for a total of 720 instances. Mentioned Branch-and-Price algorithms can solve approximately 700 out of 720 instances. The most difficult ones are those with densities in the range 0.3–0.4. The approach we propose, that is, formulating the MWSSP on graph $\hat{G}$ with the additional capacity constraints (19), can solve almost all instances with density 0.9, and a few instances with density 0.8, within a comparable computing time. However, these instances are easily solved by Branch-and-Price algorithms. Thus, we conclude that, although the method we propose can solve BPPC instances, it is not competitive with state-of-the-art Branch-and-Price algorithms.

4. Conclusions

We exploited the idea of solving Coloring problems on a graph by solving Maximum Weight Stable Set problems on an auxiliary graph, the solutions of which are in one-to-one correspondence with the colorings of the original graph, and we discuss the relation that exists between the Stable Set of the auxiliary graph and the feasible solutions of the so-called Representative formulation by Campêlo, Corrêa and Campos [4]. The contribution of the paper is twofold: first, the extensive computational experiments we performed showed that we could solve some very hard instances of the Vertex Coloring Problem on dense graphs, while for the Max-coloring Problem, the current state-of-the-art algorithm is largely outperformed; second, we extended the idea to the Equitable Coloring Problem and to the Bin Packing Problem with Conflicts, and performed extensive computational experiments for these problems as well. The experiments showed that the proposed method performs well for the Equitable Coloring Problem, while it is not competitive with state-of-the-art algorithms for the Bin Packing Problem with Conflicts. Most of the best performing algorithms for Graph Coloring problems are based on Branch-and-Price or Branch-and-Cut frameworks, which are generally tough tasks to implement and tune. Our approach is instead based on the idea of generating an effective polynomial size Mixed-Integer Programming formulation and then solving it by using a general purpose solver as a black box, exploiting in this manner the effectiveness of modern Mixed-Integer Programming solvers. The continuous development and improvement of these solvers would automatically reflect on the performances of the presented method.

Acknowledgments

The authors would like to thank Isabel Méndez-Díaz and Daniel Severín for sharing the ECP instances. The authors are grateful to two anonymous referees for their careful reading and useful comments, that helped very much in improving the paper.

References

- [1] L. Bahiense, Y. Frota, T.F. Noronha, C.C. Ribeiro, A branch-and-cut algorithm for the equitable coloring problem using a formulation by representatives, *Discrete Appl. Math.* 164 (2014) 34–46.
- [2] F. Bonomo, M. Giandomenico, F. Rossi, A note on the cornazjost transformation to solve the graph coloring problem, *Inform. Process. Lett.* 113 (18) (2013) 649–652.
- [3] D. Brélaz, New methods to color the vertices of a graph, *Commun. ACM* 22 (1979) 251–256.
- [4] M. Campêlo, V.A. Campos, R.C. Corrêa, On the asymmetric representatives formulation for the vertex coloring problem, *Discrete Appl. Math.* 156 (7) (2008) 1097–1111.
- [5] F.C. Chow, J.L. Hennessy, The priority-based coloring approach to register allocation, *ACM Trans. Program. Lang. Syst.* 12 (4) (1990) 501–536.
- [6] D. Cornaz, V. Jost, A one-to-one correspondence between colorings and stable sets, *Oper. Res. Lett.* 36 (6) (2008) 673–676.
- [7] D. Cornaz, P. Meurdesoif, Chromatic gallai identities operating on lovász number, *Math. Program.* 144 (1) (2014) 347–368.
- [8] D. de Werra, An introduction to timetabling, *European J. Oper. Res.* 19 (1985) 151–162.
- [9] S. Elhedhli, L. Li, M. Gzara, J. Naoum-Sawaya, A branch-and-price algorithm for the bin packing problem with conflicts, *INFORMS J. Comput.* 23 (3) (2011) 404–415.
- [10] E. Falkenauer, A hybrid grouping genetic algorithm, *J. Heuristics* 2 (1996) 5–30.
- [11] A.E. Fernandes-Muritiba, M. Iori, E. Malaguti, P. Toth, Algorithms for the bin packing problem with conflicts, *INFORMS J. Comput.* 22 (3) (2010) 401–415.
- [12] F. Furini, V. Gabrel, I.-C. Ternier, Lower bounding techniques for dsatur-based branch and bound, *Electron. Notes Discrete Math.* 52 (2016) 149–156.
- [13] F. Furini, E. Malaguti, Exact weighted vertex coloring via branch-and-price, *Discrete Optim.* 9 (2) (2012) 130–136.
- [14] A. Gamst, Some lower bounds for a class of frequency assignment problems, *IEEE Trans. Veh. Technol.* 35 (1) (1986) 8–14.
- [15] S. Gualandri, F. Malucelli, Exact solution of graph coloring problems via constraint programming and column generation, *INFORMS J. Comput.* 24 (1) (2012) 81–100.
- [16] S. Held, W. Cook, E.C. Sewell, Maximum-weight stable sets and safe lower bounds for graph coloring, *Math. Program. Comput.* 4 (2012) 363–381.
- [17] F.T. Leighton, A graph coloring algorithm for large scheduling problems, *J. Res. Natl. Bur. Stand.* 84 (6) (1979) 489–503.
- [18] E. Malaguti, M. Monaci, P. Toth, Models and heuristic algorithms for a weighted vertex coloring problem, *J. Heuristics* 15 (2009) 503–526.
- [19] E. Malaguti, M. Monaci, P. Toth, An exact approach for the vertex coloring problem, *Discrete Optim.* 8 (2011) 174–190.
- [20] E. Malaguti, P. Toth, A survey on vertex coloring problems, *Int. Trans. Oper. Res.* 17 (2010) 1–34.
- [21] I. Méndez-Díaz, G. Nasini, D. Severín, An exact dsatur-based algorithm for the equitable coloring problem, 2014. arXiv preprint arXiv:1405.7011.
- [22] I. Méndez-Díaz, G. Nasini, D. Severín, A polyhedral approach for the equitable coloring problem, *Discrete Appl. Math.* 164 (2014) 413–426.
- [23] I. Méndez-Díaz, P. Zabala, A branch-and-cut algorithm for graph coloring, *Discrete Appl. Math.* 154 (5) (2006) 826–847.
- [24] P. Ostergård, A fast algorithm for the maximum clique problem, *Discrete Appl. Math.* 120 (2002) 197–207.
- [25] R. Sadykov, F. Vanderbeck, Bin packing with conflicts: A generic branch-and-price algorithm, *INFORMS J. Comput.* 25 (2) (2013) 244–255.
- [26] P. San Segundo, A new dsatur-based algorithm for exact vertex coloring, *Comput. Oper. Res.* 39 (7) (2012) 1724–1733.
- [27] T.K. Woo, S.Y.W. Wu, R. Newman Wolfe, Resource allocation in a dynamically partitionable bus network using a graph coloring algorithm, *IEEE Trans. Commun.* 39 (12) (2002) 1794–1801.